

Dr Mohammad Akhlaqur Rahman

Adjunct Assistant Professor (on contract)

Department of Electronics and Communication Engineering.

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Career Objectives:

Teaching: I am eager to teach both undergraduate and graduate-level courses related to signal processing and image processing. I am specifically interested in teaching Signal and Systems, Signal processing, image processing, and courses related to artificial intelligence, deep learning and programming. I am also eager to develop new undergraduate and graduate level courses which will put special emphasis on various aspects of medical image processing, highlighting the recent trend in this area in order to foster early research ideas among students.

Research: I am interested in research areas related to detection, segmentation, and classification of various types of regions and abnormalities in medical images. This has a very wide range of application such as detection of possible signs of malignancy in radiological and pathological slide images, digital pathology, radiation dose planning for cancer treatment etc. For this I am also looking for an opportunity to collaborate with medical image processing experts, radiologists, radiation oncologists and pathologists to develop image processing-based algorithms, which can help to speed up the process of clinical examinations of radiological and pathological slides images.

Education:

Ph D	Medical Image Processing	Indian Institute of Technology Patna.	2022
M Tech	Communication and Information System.	Aligarh Muslim University	2014
B Tech	Electronics Engineering.	Aligarh Muslim University	2011

Teaching Experience:

Indian Institute of Information Technology Pune, Maharashtra.	Adjunct Assistant Professor (on contract)	12/01/2024 to Present
Sardar Vallabhbhai National Institute of Technology (SVNIT) Surat, Gujarat.	Teaching Assistant	15/03/2023 to 11/01/2024

Courses Handled:

At IIIT Pune:			
1.	CS321	Object Oriented Programming	B. Tech 2 nd Sem EC and CS Batch
At SVNIT Surat:			
2.	EC405	Deep Learning Lab	B. Tech 4th Year, 7th Semester EC
3.	EC305	Digital Signal Processing Lab	B. Tech 3rd Year, 5th Semester EC
4.	EC106	Python Programming Lab	B. Tech 1St Year, 2nd Semester EC
5.	EC104	Digital Logic Design Lab	B. Tech 1St Year, 2nd Semester EC
6.	EC101	Semiconductor Physics and Devices	B. Tech 1St Year, 1St Semester EC

List Of Publications:**Papers in Peer Reviewed Journals:**

1. SD Deb, **A Rahman**, RK Jha, “Breast cancer diagnosis using modified Xception and stacked generalization ensemble classifier,” Research on Biomedical Engineering 39 (4), 937-947, 2023.
2. **M. A. Rahman** and R. K. Jha, "Multidirectional Gabor Filter-Based Approach for Pectoral Muscle Boundary Detection," in IEEE Transactions on Radiation and Plasma Medical Sciences, vol. 6, no. 4, pp. 433-445, April 2022.
3. **M. A. Rahman**, R. K. Jha, and A. K. Gupta, “Gabor phase response-based scheme for accurate pectoral muscle boundary detection,” IET Image Processing, vol. 13, no. 5, pp. 771–778, 2019.

Conference Presentations:

1. Sundaram, G., Jha, R.K., **Rahman, M.A.**, Meena, R., Kumar, S., Kumar, P., “Biomedical Watermarking Using Arnold Transformation. In: Dhawan, A., Tripathi, V.S., Arya, K.V., Naik, K. (eds) Recent Trends in Electronics and Communication. Lecture Notes in Electrical Engineering, vol 777. Springer, Singapore. 2022.
2. Deb, Sagar Deep, **Md Akhlaqur Rahman**, and Rajib Kumar Jha,” Breast Cancer Detection and Classification using Global Pooling.” In 2020 11th International Conference on Computing, Communication and Networking Technologies (ICCCNT), pp. 1-5. IEEE, 2020.
3. **Rahman M A**, Jha R K, “Detection and Extraction of Suspected Mass Locations in Mammogram,” Research Scholar’s Day (RSD-2020), Mar. 7, 2020, IIT Patna.
4. **Rahman M A**, Jha R K, “Automatic Pectoral Muscle Boundary Detection in Mammographic Images,” Research Scholar’s Day (RSD-2019), Mar. 9, 2019, IIT Patna.
5. S Kumar, T Singh, R Jha, **MA Rahman**, “Randomness assists in wireless connectivity,” 25th International Conference on Noise and Fluctuations (ICNF 2019).
6. Sumit Kumar, R Sharma, **MA Rahman**, Rajib Kumar Jha, AR Verma, Y Singh, “Robust secret sharing for image encryption in fractional Fourier transform domain,” 2019 International Conference on Computing, Power and Communication Technologies (GUCON), New Delhi, India, 2019, pp. 497-502.

Skills:

- Programming skills: MATLAB, Python, C++, LaTeX,
- Operating Systems: UNIX / LINUX and Windows

Key Qualification and Achievement:

- GATE Qualified in 2011, 2012 and 2013. GATE-2012 (AIR-2293).